

Emmanuel Adegoke

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PROFESSIONAL SUMMARY

A Master of Engineering educated Software Engineer and Data Scientist skilled in building full-stack web applications, reproducible machine learning models, and scalable data pipelines. Demonstrates a strong career focus on translating complex multi-source datasets into actionable production solutions across software and analytics domains.

TECHNICAL SKILLS

Languages & Web: Python, SQL, R, JavaScript, React.js, HTML/CSS, REST APIs

Data Science & Machine Learning: PyTorch, TensorFlow, scikit-learn, Pandas, NumPy, SciPy, SHAP, OpenCV, Feature Engineering, Prompt Engineering

Visualisation & Analytics: Plotly, Matplotlib, Seaborn, Tableau, Power BI, Interactive Dashboards

Engineering & Infrastructure: FastAPI, Docker Containerisation, Git Version Control, PostgreSQL, Google Cloud Platform, Microsoft Fabric Concepts, Automated Testing

EXPERIENCE

Machine Learning Intern

June 2026 – Present

FlyRank (Remote)

- Built Python data pipelines to analyse 30,000 performance records, identifying 8,031 declining pages (77.6M impressions) to inform strategic optimisation.
- Trained decision tree and random forest classifiers with client-holdout validation, engineering features from search signals and engagement trends.
- Formulated a refresh-priority scoring framework to rank stale data assets, executing feature leakage audits across 18 variables prior to model training.

Student Project Lead & Data Scientist

Feb 2026 – May 2026

Structural Biology AI Project — University of Liverpool (Academic Module)

- Led an 8-person Agile team across fortnightly sprints (Jira, Git) to build a diagnostic pipeline evaluating AlphaFold protein structure predictions against ground-truth data.
- Engineered a custom rotation-invariant distance-matrix algorithm (NumPy/SciPy) and a $\mu + 2\sigma$ statistical threshold to flag geometric hallucinations in high-confidence models.
- Delivered a 9-component interactive React/Plotly dashboard, mapping prediction errors directly to structural components for research stakeholders.

PROJECTS

Predator-Prey Simulation: Multi-Agent DQN | *Python, PyTorch, Pygame, NumPy*

- Developed a multi-agent reinforcement learning environment in PyTorch, vectorising perception algorithms via NumPy broadcasting to achieve a 3–5× speedup across 650 episodes.
- Formulated a clipped reward function and resolved Q-value overestimation collapse via gradient clipping and learning rate tuning, factorising loss down to 0.9–2.4.

Football Player Recommendation System | *Python, FastAPI, Docker, React*

- Built a full-stack web application with a React frontend and FastAPI REST backend, clustering 1,000+ entities across 68 statistical features using scikit-learn K-Means and **RobustScaler**.
- Deployed the production Web API and React UI via Docker containers, achieving sub-second request-response latency from raw data ingestion to UI rendering.

Benford's Law Financial Fraud Detection | *Python, Pandas, NumPy, SciPy, Matplotlib*

- Processed and cleaned 568K transaction records in Python/Pandas, performing Chi-Square statistical hypothesis testing ($\chi^2 = 109,679$, $p \approx 0$) to uncover statistical anomalies.
- Formulated visual reports and analytical visualisations (32.63% MAPE), delivering empirical conclusions on dataset authenticity to non-technical stakeholders.

EDUCATION

University of Liverpool

Liverpool, UK

Master of Engineering in Computer Science (Honours) — 2:1

Sept 2022 – July 2026

- Relevant Modules:** Database Development (65%), Advanced Artificial Intelligence (70%), Data Mining & Visualisation (68%), Software Engineering (64%), Bioinspired Machine Learning (69%)